Canvas Quiz 5, Modules 7b–8 (graded)

Module 7b - PK-PD Relationship · Canvas quiz · 19 Oct 2026

Name _____ Date _____

Question 1 For a drug that displays linear, first-order pharmacokinetics, what is the impact on duration of activity if there is an increase in elimination half-life?

A. Duration of activity will increase.

B. There is a no change to the duration of activity.

C. Duration of activity will decrease.

Question 2 Which of the following results when the IV bolus dose of a drug displaying linear, first-order pharmacokinetics is doubled?

A. The half-life is decreased by half

B. The duration of action is increased by one half-life

C. The duration of action is doubled

D. The half-life is doubled

Question 3 What is the duration of action following a 604 mg IV bolus dose of a drug that displays linear, first-order pharmacokinetics? The half-life of elimination is 5 hours, apparent volume of distribution is 18 L, and minimum effective concentration is 6 mcg/mL. Provide your answer in hours rounded to the nearest tenth (one decimal place) with no units.

Question 4 What dose of a medication, that is to be administered as a single IV bolus injections would provide a duration of action of 20 hours in a 75 kg patient? The drug displays linear, first-order pharmacokinetics, has an elimination half-life of 5.8 hours, apparent volume of distribution of 0.25 L/kg, and a minimum effective concentration of 4 mg/L. Provide your answer in mg rounded to the nearest whole number with no units.

Question 5 Which of the following is true for a medication that displays nonlinear pharmacokinetics?

A. At low concentrations of drug in the plasma relative to Vmax, the elimination rate is independent of concentration.
B. At low concentrations of drug in the plasma relative to KM, the elimination rate depends on the concentration.
C. At low concentrations of drug in the plasma relative to Vmax, the elimination rate depends on concentration.
D. At low concentrations of drug in the plasma relative to KM, the elimination rate is independent of concentration.

Question 6 Which of the following is true regarding drugs that display nonlinear pharmacokinetics?

A. The AUC is proportional to the amount of bioavailable drug.
B. Half-life changes as dose is increased
C. Parameters do not change when different or multiple doses of a drug are given
D. The principle of superposition applies at steady-state.

Question 7 What daily dose rate of a non-linear drug would provide a steady-state concentration of 15 mg/L? In this patient, the KM is 7.9 mg/L and the Vmax is 625 mg/day. Provide your answer in mg/day rounded to the nearest five with no units.

Question 8 When a 25 year-old, 75 kg patient was taking a daily oral dose of 350 mg, his steady state plasma concentration was 8.7 mg/L.When the daily dose was increased to 450 mg, a steady-state plasma concentration of 20.6 mg/L resulted. What is the KM of this drug in this patient? Provide your answer in mg/L rounded to the nearest tenth (one decimal place) with no units.

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

Question 1 Worked by the drill

Answer: A. Duration of activity will increase.

A longer half-life means slower elimination, so the level takes longer to fall from C_0 to the MEC. teff is in direct proportion to t½: twice the half-life, twice the duration, as when decreased renal or hepatic function prolongs t½.

Question 2 Worked by the drill

Answer: B. The duration of action is increased by one half-life

Doubling D_0 doubles C_0, and the level needs exactly one half-life to fall from $2C_0$ back to C_0, after which it follows the original curve. teff rises by $\frac{0.693}{k}$, one t½; each further doubling adds one more half-life.

Question 3 Worked by the drill

Answer: 12.4 hr

WHAT THE STEM GIVES

D_0	604 mg	the numerator of C_0
t½	5 hours	gives $k = \frac{0.693}{5} = 0.1386\ hr^{-1}$
V_D	18 L	divides the dose: C_0 = 33.56 mg/L
Ceff	6 mcg/mL	equals 6 mg/L, the MEC

THE WORKING, ONE LINE AT A TIME

UNIT

$Ceff = 6\ mcg/mL = 6\ mg/L$

1 mcg/mL equals 1 mg/L, because multiplying top and bottom by 1000 turns mcg into mg and mL into L. With the dose in mg and the volume in L, C_0 then comes out in the same unit as Ceff.

SETUP

$$teff = \frac{\ln C_0 - \ln Ceff}{k}, \text{ with } C_0 = \frac{D_0}{V_D} \text{ and } k = \frac{0.693}{t½}$$

"Duration of action" after one IV bolus, with t½, V_D and the minimum effective concentration given: the natural-log duration line, the time the level takes to fall from C_0 to Ceff.

ALGEBRA

$$k = \frac{0.693}{5\ \text{hr}} = 0.1386\ \text{hr}^{-1};\ C_0 = \frac{604\ \text{mg}}{18\ \text{L}} = 33.56\ \text{mg/L}$$

The half-life gives k, the fraction of the drug eliminated per hour, in hr^{-1}. C_0 is the whole bolus spread through V_D at time zero, the level the decline starts from; mg over L is mg/L, the unit of Ceff.

ALGEBRA

$$\text{teff} = \frac{\ln(33.56 \div 6)}{0.1386\ \text{hr}^{-1}} = \frac{\ln 5.593}{0.1386} = \frac{1.721}{0.1386\ \text{hr}^{-1}} = 12.42\ \text{hr}$$

The level has to fall 5.593-fold before the effect ends; its natural log divided by k gives hours. The half-life form, 1.44 × 5 hr × 1.721 = 12.39 hr, gives the same tenth.

ROUND

12.4

Her key is 12.4 with a margin of 5%, so 11.8 to 13.0 scores. Using base-10 logs without the 2.3 gives $\frac{0.7476}{0.1386}$ = 5.39 hr, far too short; one decimal place was asked for, with no unit.

DOES THE NUMBER MAKE SENSE?

$C_0 \div$ Ceff = 5.593 is between 4 and 8, so teff is between two and three half-lives of 5 hr: 12.4 hr, between 10 and 15.

Question 4

Worked by the drill

Answer: 818 mg

WHAT THE STEM GIVES

teff	20 hours	the target duration, in the exponent
weight	75 kg	scales the volume: 0.25 × 75 = 18.75 L
t½	5.8 hours	gives k = 0.1195 hr^{-1}
V_D	0.25 L/kg	× 75 kg = 18.75 L
Ceff	4 mg/L	the MEC; times V_D gives 75 mg

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.25\ \text{L/kg} \times 75\ \text{kg} = 18.75\ \text{L};\ k = \frac{0.693}{5.8\ \text{hr}} = 0.1195\ \text{hr}^{-1}$$

The volume is given per kilogram, so it is scaled to the 75 kg patient first. The dose line has k in its exponent, so the half-life is turned into k.

SETUP

$$\text{Ceff} = \frac{D_0}{V_D}\, e^{-k\ \text{teff}},\ \text{so}\ D_0 = \text{Ceff} \times V_D \times e^{k\ \text{teff}}$$

$C = C_0 e^{-kt}$ with $C_0 = D_0$ over V_D, written at the moment the level reaches the MEC, 20 hours after the dose, then solved for the dose.

ALGEBRA

$D_0 = (4\ \text{mg/L})(18.75\ \text{L})\ e^{(0.1195)(20)} = 75\ \text{mg} \times e^{2.390} = 75\ \text{mg} \times 10.91 = 818.2\ \text{mg}$

mg/L × L leaves mg: 75 mg is the amount in the body when the level is at the MEC. k teff has no units, and the dose must be $e^{2.390} = 10.91$ times that 75 mg.

ROUND

818

Her key is 818 with a margin of 5%, so 777 to 859 scores. With ln 2 in place of 0.693 the dose is 818.6 mg, which rounds to 819 and also scores. Dividing by the exponential instead gives $\frac{75}{10.91} = 6.9$ mg, far too small.

DOES THE NUMBER MAKE SENSE?

20 hr is between 3 and 4 half-lives of 5.8 hr, and each half-life doubles the dose, so the dose is between 8 and 16 times the 75 mg at the MEC: 818 mg.

Question 5

Worked by the drill

Answer: B. At low concentrations of drug in the plasma relative to KM, the elimination rate depends on the concentration.

With C_p much less than KM, C_p is negligible in KM + C_p and the rate is $\frac{Vmax}{KM} \times C_p = k'C_p$. Vmax and KM are constants, so the rate is proportional to C_p: first-order behaviour.

Question 6

Worked by the drill

Answer: B. Half-life changes as dose is increased

With saturable elimination $t_{½} = V_D(\frac{0.693}{Vmax})(KM + C_p)$, so a larger dose, giving a higher C_p, gives a longer half-life. It is one of the characteristics of saturation kinetics.

Question 7

Worked by the drill

Answer: 409 mg/day

WHAT THE STEM GIVES

C_{ss}	15 mg/L	the target, top and bottom
KM	7.9 mg/L	added to C_{ss} in the denominator
Vmax	625 mg/day	the numerator; units of R

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{Vmax \times C_{ss}}{KM + C_{ss}}$$

KM and Vmax are given, and the target C_{ss} is 15 mg/L, so the nonlinear dose-rate line answers directly. Vmax is already in mg/day, so no weight is needed.

ALGEBRA

$$R = \frac{(625\ \text{mg/day})(15\ \text{mg/L})}{(7.9 + 15)\ \text{mg/L}} = \frac{9375}{22.9} = 409.4\ \text{mg/day}$$

mg/L cancels top and bottom and leaves mg/day, the units of Vmax. The target C_{ss} is on top and again, added to KM, on the bottom, which keeps R below Vmax for any target.

ROUND

$R = 409$ mg/day

Her key is 409, the whole number; the instruction asks for the nearest five, which is 410. The margin of 5% takes 389 to 430, so both score. Leaving KM out of the bottom gives Vmax itself, 625, far too high.

> **DOES THE NUMBER MAKE SENSE?**
> C_{ss} is about two-thirds of KM + C_{ss} (15 of 22.9), so R is about two-thirds of 625: near 410 mg/day, below Vmax.

Canvas shows 409 as the key, although the stem asks for the nearest five (410). Both are within the ±5% margin, so either scores here.

Question 8

Worked by the drill

Answer: 5.4 mg/L

WHAT THE STEM GIVES

R1	350 mg	the first daily dose
C_1	8.7 mg/L	steady state on 350 mg
R2	450 mg	the increased daily dose
C_2	20.6 mg/L	steady state on 450 mg
weight	75 kg	*not needed: the doses stay in mg/day*
age	25 year-old	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$$, with R1 = 350 mg/day, C_1 = 8.7 mg/L, R2 = 450 mg/day, C_2 = 20.6 mg/L

Two daily doses in the same patient and the steady-state concentration each produced: the direct method. The 75 kg weight and the age are not needed when the doses stay in mg/day.

ALGEBRA

$$KM = \frac{450 - 350}{(350/8.7) - (450/20.6)} = \frac{100}{40.23 - 21.84} = \frac{100}{18.39} = 5.438\ \text{mg/L}$$

Each dose over its own concentration: 350 over 8.7 is 40.23 and 450 over 20.6 is 21.84. (mg/day) over (mg/day per mg/L) leaves mg/L, a concentration, the unit of KM.

ROUND

KM = 5.4 mg/L

Her key is 5.4 with a margin of 5%, so 5.13 to 5.67 scores. Pairing each dose with the other dose's concentration makes the bottom negative and gives a negative KM, which cannot be right.

DOES THE NUMBER MAKE SENSE?

Raising the dose from 350 to 450 mg more than doubled the concentration, from 8.7 to 20.6 mg/L, the mark of saturation; KM is a concentration: 5.4 mg/L.